

## GANGNEUNG AB, Korea, Republic of

WMO: 471070

Lat: 37.754N

Lon: 128.944E

Elev: 35

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 82-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 15.8       | 18.6 | -1.9                          | 5.0 | 24.8 | 3.0 | 6.4 | 25.0 | 29.0                  | 34.2 | 26.6 | 31.7 | 10.2                  | 260  | 0.421 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 12.3                      | 90.5            | 77.0 | 87.8 | 76.3 | 84.6 | 74.2 | 79.2                | 86.8 | 78.0 | 85.0 | 76.7 | 83.1 | 9.3                     | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.1                            | 141.5 | 83.4 | 75.6 | 134.0 | 81.4 | 74.9 | 131.1 | 80.7 | 42.8          | 86.5 | 41.4 | 85.6 | 40.1 | 83.7 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 22.8              | 19.0 | 16.7 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |
|                   |      |      | 8.0                        | 83.5 | 2.8                | 2.2 | 6.0                                                | 85.1 | 4.4        | 86.4 | 2.9        | 87.6 | 0.9        | 89.2 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 54.4  | 31.6  | 34.2  | 41.9  | 53.0  | 61.6  | 67.7  | 75.0  | 76.0  | 67.3  | 57.5  | 47.5  | 38.5 | (6)  |
| (7)  |                                                                   | DBStd       | 16.12 | 6.15  | 6.04  | 6.03  | 7.18  | 6.30  | 5.60  | 6.10  | 4.84  | 4.92  | 5.63  | 6.35  | 6.53 | (7)  |
| (8)  |                                                                   | HDD50       | 1816  | 572   | 443   | 261   | 47    | 1     | 0     | 0     | 0     | 0     | 11    | 119   | 362  | (8)  |
| (9)  |                                                                   | HDD65       | 4774  | 1037  | 863   | 716   | 366   | 143   | 30    | 4     | 0     | 29    | 239   | 525   | 822  | (9)  |
| (10) |                                                                   | CDD50       | 3431  | 0     | 0     | 10    | 137   | 361   | 531   | 776   | 806   | 518   | 242   | 45    | 5    | (10) |
| (11) |                                                                   | CDD65       | 915   | 0     | 0     | 0     | 6     | 39    | 111   | 315   | 341   | 98    | 5     | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 6775  | 0     | 0     | 0     | 50    | 312   | 733   | 2565  | 2563  | 510   | 40    | 2     | 0    | (12) |
| (13) |                                                                   | CDH80       | 2117  | 0     | 0     | 0     | 8     | 55    | 194   | 920   | 842   | 94    | 4     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 6.3   | 9.1   | 7.1   | 7.4   | 7.4   | 6.2   | 4.9   | 5.2   | 4.3   | 4.4   | 5.2   | 6.5   | 8.2  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 55.2  | 2.0   | 2.1   | 2.8   | 3.1   | 3.1   | 4.4   | 9.1   | 10.9  | 8.7   | 4.2   | 3.1   | 1.5  | (15) |
| (16) |                                                                   | PrecMax     | 82.5  | 9.8   | 7.7   | 6.9   | 9.8   | 10.9  | 13.5  | 29.8  | 44.8  | 22.0  | 13.7  | 13.7  | 9.2  | (16) |
| (17) |                                                                   | PrecMin     | 36.4  | 0.0   | 0.0   | 0.3   | 0.0   | 0.1   | 0.6   | 2.3   | 2.2   | 0.0   | 0.1   | 0.0   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 11.0  | 2.0   | 1.9   | 1.7   | 2.2   | 2.2   | 2.8   | 5.3   | 7.0   | 5.9   | 3.5   | 3.0   | 1.8  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 51.7  | 55.9  | 66.6  | 80.6  | 85.6  | 90.9  | 93.6  | 95.0  | 87.4  | 78.7  | 71.5  | 60.9 | (19) |
| (20) |                                                                   | MCWB        | 43.0  | 46.7  | 50.1  | 63.0  | 65.2  | 72.0  | 78.7  | 80.8  | 75.4  | 63.0  | 56.6  | 50.1  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 48.1  | 50.3  | 60.7  | 73.8  | 80.9  | 85.6  | 90.0  | 91.2  | 82.5  | 73.6  | 65.5  | 57.1 | (21) |
| (22) |                                                                   | MCWB        | 40.4  | 42.0  | 47.6  | 57.1  | 63.3  | 69.6  | 77.3  | 78.4  | 72.8  | 62.2  | 54.7  | 47.0  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 44.9  | 46.7  | 56.3  | 69.9  | 77.7  | 82.1  | 87.8  | 87.9  | 79.0  | 71.4  | 62.5  | 53.5 | (23) |
| (24) |                                                                   | MCWB        | 38.0  | 39.6  | 46.1  | 55.3  | 62.2  | 68.9  | 76.9  | 77.5  | 70.4  | 61.9  | 53.3  | 44.0  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 42.5  | 44.3  | 53.1  | 66.3  | 74.2  | 78.6  | 85.5  | 84.5  | 76.7  | 68.4  | 59.2  | 50.1 | (25) |
| (26) |                                                                   | MCWB        | 36.2  | 38.5  | 44.3  | 53.7  | 60.7  | 67.6  | 75.6  | 76.2  | 69.3  | 59.9  | 50.8  | 41.7  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 44.1  | 48.0  | 52.1  | 65.1  | 69.8  | 75.3  | 80.7  | 81.8  | 77.4  | 67.6  | 59.2  | 51.5 | (27) |
| (28) |                                                                   | MADB        | 50.3  | 52.1  | 62.1  | 74.7  | 79.0  | 84.3  | 88.7  | 91.5  | 85.0  | 73.2  | 65.5  | 58.4  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 41.1  | 43.7  | 49.7  | 60.6  | 66.2  | 72.5  | 79.1  | 79.8  | 74.7  | 65.4  | 56.7  | 48.1 | (29) |
| (30) |                                                                   | MADB        | 46.6  | 47.9  | 58.1  | 69.8  | 77.7  | 80.7  | 87.2  | 87.6  | 80.1  | 70.8  | 63.8  | 56.3  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 38.9  | 40.8  | 47.1  | 57.4  | 63.9  | 70.6  | 77.8  | 78.5  | 72.5  | 63.5  | 54.5  | 44.9 | (31) |
| (32) |                                                                   | MADB        | 44.5  | 45.3  | 54.0  | 66.6  | 74.4  | 78.0  | 85.7  | 85.1  | 77.6  | 68.9  | 60.7  | 51.1  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 36.4  | 38.8  | 44.9  | 54.9  | 61.9  | 69.0  | 76.5  | 77.3  | 70.2  | 61.1  | 52.2  | 42.7 | (33) |
| (34) |                                                                   | MADB        | 41.5  | 43.1  | 51.5  | 63.8  | 71.0  | 75.9  | 83.5  | 83.4  | 74.9  | 67.0  | 58.0  | 49.3  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 16.8  | 15.0  | 16.0  | 17.8  | 17.0  | 12.9  | 11.6  | 12.3  | 15.8  | 19.3  | 18.6  | 16.5  | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 18.8  | 18.5  | 21.5  | 23.0  | 23.7  | 19.8  | 16.5  | 17.7  | 18.5  | 22.3  | 22.4  | 20.3 | (36) |
| (37) |                                                                   | MCWBR       | 14.3  | 13.2  | 14.0  | 14.3  | 13.5  | 10.3  | 8.5   | 8.7   | 11.3  | 14.5  | 15.3  | 14.2  | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 18.4  | 16.3  | 19.9  | 20.6  | 20.8  | 16.6  | 14.7  | 15.0  | 16.0  | 19.5  | 21.1  | 18.5 | (38) |
| (39) |                                                                   | MCWBR       | 15.0  | 12.5  | 14.3  | 14.3  | 13.1  | 10.2  | 8.6   | 8.5   | 10.8  | 14.2  | 15.6  | 14.2  | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.422 | 0.487 | 0.584 | 0.625 | 0.630 | 0.680 | 0.641 | 0.571 | 0.498 | 0.490 | 0.454 | 0.413 | (40) |      |
| (41) |                                                                   | taud        | 1.966 | 1.842 | 1.681 | 1.651 | 1.684 | 1.647 | 1.788 | 1.962 | 2.088 | 1.978 | 1.997 | 2.019 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 225   | 226   | 216   | 218   | 220   | 209   | 216   | 229   | 239   | 223   | 213   | 218   | (42) |      |
| (43) |                                                                   | Edh at Noon | 43    | 56    | 72    | 78    | 77    | 80    | 69    | 56    | 47    | 48    | 42    | 38    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 758   | 1005  | 1303  | 1556  | 1724  | 1637  | 1367  | 1359  | 1236  | 1078  | 764   | 679   | (44) |      |
| (45) |                                                                   | RadStd      | 65    | 96    | 128   | 121   | 169   | 138   | 192   | 169   | 148   | 94    | 99    | 57    | (45) |      |

## Historical Trends

|                         |     | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|-------------------------|-----|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                         |     |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)                     | (b) | (c)   | (d)     | (e)    | (f)   | (g)     | (h)   | (i) | (j)         | (k)   | (l)   | (m)   |
| Station Only            |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| Regional (34 neighbors) |     |       | N/A     | N/A    | -1.69 | N/A     | N/A   | N/A | N/A         | N/A   | +175  | +94   |

Nomenclature: See separate page